

第1次作业总结

题3

题目

Determine the frequency counts for all statements in the following two pro-gram segments:

(a)

```
1  for(i=1; i ≤ n; i++)
2      for(j=1; j ≤ i; j++)
3          for(k=1; k ≤ j; k++)
4              x++;
```

(b)

```
1  i=1;
2  while(i ≤ n)
3  {
4      x++;
5      i++;
6  }
```

参考答案

(a)

line 代码	frequency
1 for(i=1; i ≤ n; i++)	$n + 1$
2 for(j=1; j ≤ i; j++)	$(1 + 1) + (2 + 1) + \dots + (n + 1) = \frac{1}{2}n(n + 3)$
3 for(k=1; k ≤ j; k++)	$2 + (2 + 3) + \dots + (2 + 3 + \dots + n + 1) = \frac{n(n+1)(n+5)}{6}$
4 x++;	$1 + (1 + 2) + \dots + (1 + 2 + \dots + n) = \frac{n(n+1)(n+2)}{6}$

(b)

line	代码	frequency	total steps
1	i=1;	1	1
2	while(i<=n)	$n + 1$	$n + 1$
3	{	1	0
4	x++;	n	n
5	i++;	n	n
6	}	1	0

主要出错 (a) 的 line3 和 line4

题4

题目

(a) Introduce statements to increment count at all appropriate points in Program 1.32.

```

1  void D(int *x, int n){
2      int i = 1;
3      do {
4          x[i] += 2;
5          i+= 2;
6      }
7      while (i ≤ n);
8      i = 1;
9      while (i ≤ (n/2))
10     {
11         x[i] += x[i+1];
12         i++;
13     }
14 }
15 // Program 1.32:Example program

```

(b) Simplify the resulting program by eliminating statements. The simplified program should compute the same value for count as computed by the program of (a).

(c) What is the exact value of count when the program terminates? You may assume that the initial value of count is 0.

(d) Obtain the step count for Program 1.32 using the frequency method. Clearly show the step count table.

参考答案

(a)

```
1 void D(int *x, int n){
2     int i = 1; count++;
3     do {
4         x[i] += 2; count++;
5         i+= 2; count++;
6         count++;
7     }
8     while (i ≤ n);
9     i = 1; count++;
10    while (i ≤ (n/2))
11    {
12        count++;
13        x[i] += x[i+1]; count++;
14        i++; count++;
15    }
16    count++;
17 }
```

(b)

```
1 void D(int *x, int n){
2     int i = 1; count++;
3     do {
4         count++;
5         i+= 2; count++;
6         count++;
7     }
8     while (i ≤ n);
9     i = 1; count++;
10    while (i ≤ (n/2))
11    {
12        count++;
13        count++;
14        i++; count++;
15    }
16    count++;
17 }
```

(c)

$$n \text{ 为奇数: } 1 + 3 * \frac{n+1}{2} + 3 * \frac{n-1}{2} + 1 = 3n + 2$$

$$n \text{ 为偶数: } 1 + 3 * \frac{n}{2} + 3 * \frac{n}{2} + 1 = 3n + 2$$

因此: $counts = 3n + 2$

(d)

line	s/e	frequency (n为奇数/n为偶数)	total steps (n为奇数/n为偶数)
2	1	1	1
3	0	$\frac{n+1}{2} / \frac{n}{2}$	0
4	1	$\frac{n+1}{2} / \frac{n}{2}$	$\frac{n+1}{2} / \frac{n}{2}$
5	1	$\frac{n+1}{2} / \frac{n}{2}$	$\frac{n+1}{2} / \frac{n}{2}$
6	0	$\frac{n+1}{2} / \frac{n}{2}$	0
7	1	$\frac{n+1}{2} / \frac{n}{2}$	$\frac{n+1}{2} / \frac{n}{2}$
8	1	1	1
9	1	$\frac{n+1}{2} / \frac{n}{2} + 1$	$\frac{n+1}{2} / \frac{n}{2} + 1$
10	0	$\frac{n-1}{2} / \frac{n}{2}$	0
11	1	$\frac{n-1}{2} / \frac{n}{2}$	$\frac{n-1}{2} / \frac{n}{2}$
12	1	$\frac{n-1}{2} / \frac{n}{2}$	$\frac{n-1}{2} / \frac{n}{2}$
13	0	$\frac{n-1}{2} / \frac{n}{2}$	0

主要出错在 (c) 和 (d)，要分为n是奇数和偶数两种情况考虑。

题5

题目

Do Exercise 4 for function Transpose (Program 1.33)

```
1 void Transpose(int **a, int n)
2 {
3     for(int i=0;i<n-1;i++)
4         for(int j=i+1;j<n;j++)
5             swap(a[i][j],a[j][i]);
6 }
7 // Program 1.33: Matrix transpose
```

参考答案

(a)

```
1 void Transpose(int **a, int n)
2 {
3     for(int i=0;i<n-1;i++)
4     {
5         count++;
6         for(int j=i+1;j<n;j++)
7         {
8             count++;
9             swap(a[i][j],a[j][i]);
10            count+=3;
11        }
12        count++;
13    }
14    count++;
15 }
```

(b)

```
1 void Transpose(int **a, int n)
2 {
3     for(int i=0;i<n-1;i++)
4     {
5         count++;
6         for(int j=i+1;j<n;j++)
7         {
8             count++;
9             count+=3;
10        }
11        count++;
12    }
13    count++;
14 }
```

(c)

$$counts = n - 1 + 4 * (1 + 2 + \dots + n - 1) + n - 1 + 1 = 2n^2 - 1$$

(d)

line	s/e	frequency	total steps
3	1	n	n
4	1	$n + \dots + 2 = \frac{(n+2)(n-1)}{2}$	$\frac{(n+2)(n-1)}{2}$
5	3	$(n-1) + \dots + 1 = \frac{n(n-1)}{2}$	$\frac{3n(n-1)}{2}$

在计算 *s/e* 时，有的同学将 *swap()* 一句计为 3，有的同学计为 1，都可以。*frequency* 的计算也和第 3 题一样，进入 *for* 循环多少次会容易出错。

题6

题目

Do Exercise 4 for Program 1.34. This program multiplies two $n \times n$ matrices *a* and *b*.

```
1 void Multiply(int **a, int **b, int **c, int n)
2 {
3     for(int i=0; i<n; i++)
4         for(int j=0; j<n; j++)
5             {
6                 c[i][j]=0;
7                 for(int k=0; k<n; k++)
8                     c[i][j] += a[i][k] * b[k][j];
9             }
10 }
11 // Program 1.34: Square matrix multiplication
```

参考答案

(a)

```
1 void Multiply(int **a, int **b, int **c, int n)
2 {
3     for(int i=0; i<n; i++)
4     {
5         count++;
6         for(int j=0; j<n; j++)
7         {
8             count++;
```

```

9         c[i][j]=0;
10        count++;
11        for(int k=0;k<n;k++){
12            count++;
13            c[i][j]+=a[i][k]*b[k][j];
14            count++;
15        }
16        count++;
17    }
18    count++;
19 }
20 count++;
21 }

```

(b)

```

1 void Multiply(int **a,int **b,int **c,int n)
2 {
3     for(int i=0;i<n;i++)
4     {
5         for(int j=0;j<n;j++)
6         {
7             for(int k=0;k<n;k++){
8                 count+=2;
9             }
10            count+=3;
11        }
12        count+=2;
13    }
14    count++;
15 }

```

(c)

$$counts = 2n^3 + 3n^2 + 2n + 1$$

(d)

line	s/e	frequency	total steps
1	/	/	/
3	1	$n + 1$	$n + 1$


```

9         c[i][j]=0;
10        count++;    // count for assignment
11        for(int k=0;k<n;k++)
12        {
13            count++;    // count for "for_3"
14            c[i][j]+=a[i][k]*b[k][j];
15            count++;    // count for assignment
16        }
17        count++;    // count for the last "for_3"
18    }
19    count++;    // count for the last "for_2"
20 }
21 count++;    // count for the last "for_1"
22 }

```

(a) (b)

```

1 void Multiply(int **a,int **b,int **c,int m,int n,int p)
2 {
3     for(int i=0;i<m;i++)
4     {
5         count+=2;
6         for(int j=0;j<p;j++)
7         {
8             count+=3;
9             for(int k=0;k<n;k++)
10            {
11                count+=2;
12            }
13        }
14    }
15    count++;
16 }

```

(a) (c)

$$counts = 2mpn + 3mp + 2m + 1$$

(a) (d)

line	s/e	frequency	total steps
1	/	/	/

line	s/e	frequency	total steps
2	0	1	0
3	1	$m + 1$	$m + 1$
4	0	m	0
5	1	$m * (p + 1)$	$m * (p + 1)$
6	0	$m * p$	0
7	1	$m * p$	$m * p$
8	1	$m * p * (n + 1)$	$m * p * (n + 1)$
9	0	$m * p * n$	0
10	1	$m * p * n$	$m * p * n$
11	0	$m * p * n$	0
12	0	$m * p$	0
13	0	m	0
14	0	1	0

(b)

solution: when p is smaller than m , it is profitable to interchange the two outermost "For" loops.